Personalised Prediction of Postprandial Glucose Responses from Meal, Activity, and Metabolic Data

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Abstract

Background. Blood glucose responses to the same meal vary widely between people, which limits generic dietary advice.

Objective. To test whether meal, activity and metabolic data predict an individual's post-meal glucose response better than carbohydrate content, whether personalisation with a person's own past responses helps, and whether accuracy holds across metabolic groups.

Methods. I used CGMacros: 45 adults (15 healthy, 16 prediabetes, 14 type 2 diabetes) with about 10 days of continuous glucose monitoring and 1,403 analysable meals. Targets were the glucose change at 30, 60 and 120 minutes, the 2-hour incremental area under the curve (iAUC), the full 2-hour curve, and high-spike detection (peak rise of 50 mg/dL or more). I compared baselines, LightGBM with and without personal history features, and a temporal convolutional network with per-person fine-tuning. Evaluation used cross-validation grouped by participant, with participant-level bootstrap confidence intervals.

Results. Without personalisation, LightGBM with history features reduced 60-minute error by 24% versus a carbs-only model (95% CI 18–30%). Personalised LightGBM, trained with each person's earlier meals, reduced it by 38% (17.8 vs 28.6 mg/dL; 95% CI 32–44%) and reached r = 0.78 for iAUC. Averaging personalised LightGBM and the fine-tuned network gave a spike-detection AUROC of 0.89. Personalisation cut type 2 diabetes error by about 40%, but a gap with healthy participants remained. Results held across sensors and data-cleaning rules.

Conclusion. A person's own response history mattered more than model architecture. With 45 people and no microbiome data, personalised models reached accuracy in the range of much larger published studies.

1. Introduction

Post-meal (postprandial) glucose spikes are linked to the progression of type 2 diabetes and to cardiovascular risk. Dietary advice usually relies on carbohydrate counting or food-level scores such as the glycemic index, which assume that a given food affects everyone similarly.

Large studies have shown this assumption is wrong: people's responses to identical meals differ widely, so models that use personal information predict responses far better than carbohydrate content alone. Those studies used hundreds of participants and gut microbiome profiling, which is expensive and not routinely available.

This project asks whether useful personalised prediction is possible with a small cohort, routine wearable data and standard blood tests. It addresses four questions:

1. **RQ1.** Do meal, timing, activity and metabolic features predict an individual's glucose response better than carbohydrate content?
2. **RQ2.** Does using a person's own earlier meal responses (personalisation) improve prediction?
3. **RQ3.** Does a temporal deep learning model outperform gradient-boosted trees on this task?
4. **RQ4.** Does accuracy hold across healthy, prediabetic and type 2 diabetic participants, and across sex and age?

The focus is glycemic **response** (how one person's glucose changes after one meal), not the glycemic **index** (a fixed score per food).

2. Related Work

Three studies frame this project. Each found that personal factors predict glucose responses far better than carbohydrate content, and they differ on whether a person's own data is needed.

Study	Participants	Inputs	Key result
Zeevi et al. 2015, *Cell*	800 Israeli adults (46,898 meals), validated on 100	Blood tests, diet, anthropometrics, activity, gut microbiome	Gradient boosting predicted individual iAUC far better than carbohydrate or calorie counting (validation correlation about 0.7; to verify)
Berry et al. 2020, *Nature Medicine*	1,002 UK adults (PREDICT 1), validated on 100 US adults	Meals, microbiome, genetics, blood tests	Glucose response varied 68% between people; macronutrients explained 15.4% of variance, microbiome 6.0%; model r = 0.77

Study	Participants	Inputs	Key result
Shen, Choi & Kleinberg, *J Diabetes Sci Technol*	397 with T1D, 100 with T2D	Diet, demographics, timing; no microbiome	r = 0.61–0.73; food categories beat macronutrients; personalised training data was not required

CGMacros (Das et al., *Scientific Data*; PhysioNet, 2025) is a public pilot dataset covering the full metabolic spectrum, from healthy to type 2 diabetes. It combines two CGM sensors, wearable activity, meal macronutrients and blood tests, which makes it suited to studying personalisation in a small, mixed cohort.

Gap addressed. The large studies relied on hundreds of participants and microbiome data, and Shen et al. studied people who already had diabetes. This project tests personalisation, model choice and subgroup fairness together in a small cohort spanning healthy to type 2 diabetes, using only routine data.

3. Data

The analysis uses 1,403 meals from all 45 CGMacros participants (v1.0.0, open access, CC BY-NC-SA 4.0).

Cohort. 29 women and 16 men, aged 18–69, mostly Hispanic/Latino (34 of 45). Participants were grouped by baseline HbA1c using ADA cut-offs: below 5.7% healthy (15), 5.7–6.4% prediabetes (16), 6.5% or above type 2 diabetes (14). This reproduces the dataset's published group sizes.

Signals.

- **Glucose:** Abbott Libre Pro (primary, 1-minute resolution, available for every participant) and Dexcom G6 Pro (used in the sensitivity analysis).
- **Activity:** Fitbit heart rate, calories burned and METs per minute.
- **Meals:** meal type, time, estimated calories, carbohydrate, protein, fat, fiber and percentage eaten; 1,706 meals and snacks were logged.
- **Blood tests:** baseline HbA1c, fasting glucose, fasting insulin, triglycerides and HDL, plus age, sex and BMI.

Exclusions (1,706 → 1,403 meals).

Rule	Meals removed	Reason
Another meal or snack within 2 hours	250	Overlapping responses can't be attributed to one meal
Implausible log (over 3,000 kcal, or macronutrient energy outside 0.5–2× logged calories)	46	Clear logging errors, e.g. 2,283 g carbohydrate in one dinner
CGM gap over 20 minutes in the 2-hour window	6	Gaps are not invented by interpolation
Missing pre-meal glucose trace	1	Needed by the temporal model

Sensor gaps of 20 minutes or less were filled by linear interpolation. Longer gaps were left missing.

4. Methods

4.1 Targets

For each meal: glucose change from meal start at 30, 60 and 120 minutes; 2-hour iAUC (area above the starting level, mg/dL·min); peak rise; the full curve at 5-minute steps (24 points); and a binary high spike (peak rise of 50 mg/dL or more, which occurred after 34% of meals).

4.2 Features

All features are known at or before the start of the meal.

- **Meal:** carbohydrate, protein, fat, fiber and calories, scaled by the percentage eaten.
- **Timing:** hour of day (sine and cosine) and minutes since the previous meal.
- **Current state:** glucose at meal start, its 30-minute slope, and the 60-minute pre-meal trace (temporal model only).
- **Activity:** calories burned, mean METs and mean heart rate in the hour before the meal.
- **Person:** age, sex, BMI, HbA1c, fasting glucose, fasting insulin, triglycerides, HDL.
- **Personal history:** computed only from the same person's earlier meals (at least 3): mean past 60-minute rise, mean past iAUC and peak, median past rise per 10 g carbohydrate, and the previous meal's rise.

4.3 Models

ID	Model	Personalised
B0	Mean response of training participants in the same metabolic group	No
B1	Linear regression on carbohydrate only	No
M1	LightGBM on meal, timing, activity and person features	No
M2	M1 plus personal history features	Partly
M2b	M2, also trained on the test person's first half of meals	Yes
M3	Temporal CNN predicting the full 2-hour curve	Partly
M4	M3 with the output head fine-tuned on the test person's first half of meals	Yes

LightGBM used fixed settings (300 trees, learning rate 0.03, 15 leaves), one model per target. **The temporal CNN** applies three dilated 1-D convolutions to the pre-meal glucose trace, fuses the result with the other features through a small MLP, and outputs 24 curve points. It was trained with L1 loss as a 3-seed ensemble. Fine-tuning (M4) updates only the output head, with a low learning rate and strong weight decay, to avoid overfitting about 15 meals per person.

4.4 Evaluation

- **New-people setting:** stratified 5-fold cross-validation grouped by participant, so every test person is unseen during training.
- **Personalised setting:** within the same folds, each test person's chronologically earlier half of meals is used for adaptation, and all models are scored on the same later half (709 meals).
- **Metrics:** mean absolute error (MAE); Pearson r for comparison with published work; AUROC, sensitivity and specificity for spike detection.
- **Uncertainty:** 95% confidence intervals from 1,000 bootstrap resamples of participants (not meals), since meals from one person are correlated.
- **No hyperparameter tuning**, to avoid optimistic estimates on a small cohort.
- **Leakage controls:** pre-meal features only, history computed with a one-meal lag, and feature scaling fitted on training data only.

5. Results

5.1 New people (RQ1, RQ3)

Adding history features gave the most accurate model for unseen people: 60-minute MAE fell from 28.8 mg/dL (carbs only) to 22.0, a 24% reduction (95% CI 18–30%). History alone added 11% over M1 (95% CI 5–18%).

Model	30 min	60 min	120 min	iAUC (mg/dL·min)
B0 Group mean	18.7	28.0	27.2	2036
B1 Carbs only	18.6	28.8	28.0	2141
M1 LightGBM	16.0	24.8	24.3	1697
M2 LightGBM + history	14.1	**22.0**	**23.7**	**1600**
M3 Temporal CNN	**13.1**	24.1	25.4	1653

MAE in mg/dL unless stated; 1,403 meals. The CNN was best at 30 minutes, but worse than M2 at 60 minutes (the difference was not clear: 95% CI −1% to +27% error).

5.2 Personalised prediction (RQ2, RQ3)

Personalisation produced the largest gain in the study. On each person's later meals, personalised LightGBM reduced 60-minute error by 38% versus carbs only (95% CI 32–44%).

Model	30 min	60 min	120 min	iAUC	Full curve
B1 Carbs only	18.4	28.6	26.9	2107	—
M2 LightGBM + history	13.5	20.9	23.3	1551	—
M3 Temporal CNN	12.7	23.4	24.4	1602	19.0
M4 CNN fine-tuned	**10.7**	19.1	19.9	**1260**	**15.4**
M2b LightGBM personalised	12.2	**17.8**	**18.7**	1269	—

709 later-half meals. Fine-tuning improved the CNN by 18% at 60 minutes (95% CI 9–29%). M2b was 7% more accurate than M4 at 60 minutes (95% CI 1–12%), while M4 was best at 30 minutes and for the full curve.

5.3 Spike detection

Averaging the two personalised models gave the best AUROC, 0.89 (95% CI 0.86–0.92). Carbohydrate content alone was a weak detector (0.60).

Model	Setting	AUROC (95% CI)	Sensitivity	Specificity
B1 Carbs only	New people	0.60 (0.53–0.66)	9%	94%
M2 LightGBM + history	New people	0.82 (0.78–0.85)	62%	85%
M2b LightGBM personalised	Personalised	0.87 (0.84–0.90)	67%	86%
M4 CNN fine-tuned	Personalised	0.87 (0.84–0.91)	55%	93%
Average of M2b and M4	Personalised	**0.89 (0.86–0.92)**	—	—

The average used equal-weight ranks with no tuning. It was the best of three combinations tried, so 0.89 should be read as approximate. A dedicated spike classifier (0.86) did not beat thresholding the predicted peak.

5.4 Subgroups (RQ4)

History features helped most where the base model was weakest: type 2 diabetes and men.

Subgroup	People / meals	Carbs only	M1	M2 (95% CI)
Healthy	15 / 472	23.9	18.0	16.9 (13.3–22.8)
Prediabetes	16 / 519	26.5	22.2	21.2 (18.4–24.4)
Type 2 diabetes	14 / 412	37.1	35.9	28.7 (24.6–33.9)
Female	29 / 905	27.4	22.9	21.0 (18.1–24.3)
Male	16 / 498	31.2	28.2	23.7 (18.9–29.4)
Age under 40	10 / 333	20.9	15.0	14.9 (12.9–17.3)
Age 40+	35 / 1070	31.2	27.8	24.2 (20.9–27.6)

60-minute MAE, mg/dL, new-people setting. With personalisation (M2b), error was 15.8, 16.5 and 21.7 mg/dL for healthy, prediabetes and type 2 diabetes. Type 2 diabetes error remained about 12 mg/dL above healthy (95% CI 4–19).

5.5 Robustness

The improvement over carbs only stayed at 23–25% across every variant: Dexcom instead of Libre (24%), no overlap exclusion (25%) and no plausibility filter (24%). Absolute errors were higher with Dexcom (24.8 vs 22.1 mg/dL), but model rankings never changed.

5.6 Drivers of predicted spikes

SHAP values for the non-personalised 60-minute model matched known physiology:

- Higher fasting glucose, carbohydrate and HbA1c raised the predicted rise.
- More protein and fiber lowered it.
- A higher glucose level at meal start predicted a smaller further rise.
- More activity in the preceding hour slightly lowered it.

5.7 Comparison with published work

For iAUC, M2 reached r = 0.63 for new people and M2b reached r = 0.78 when personalised, compared with r = 0.12 for carbs only. This places the personalised model in the range of Berry et al. (0.77) and Shen et al. (0.61–0.73), with 45 participants and no microbiome data. The comparison is indicative, since cohorts, meals and validation designs differ.

6. Discussion

Personalisation mattered more than model choice. The step from carbs only to a non-personalised LightGBM cut 60-minute error by 14%. Adding a person's history and a few days of their own meals raised the reduction to 38%. In contrast, the gap between the best tree model and the best deep model was 7%. For a clinic or app, collecting a short run of CGM data per person is likely worth more than a more complex model.

Trees and deep models were complementary, not competing. With about 1,400 meals, LightGBM was more accurate at 60 and 120 minutes, which matches the common finding that tree ensembles are strong on small tabular data. The temporal CNN was better at 30 minutes, where the pre-meal glucose trend carries most of the signal, and it is the only model that outputs a full curve. Averaging the two improved spike detection, which suggests their errors differ.

Personalisation narrowed the type 2 diabetes gap but did not close it. The non-personalised model barely beat carbs only for type 2 diabetes (35.9 vs 37.1 mg/dL). History features cut this to 28.7, and personalisation to 21.7. Responses in type 2 diabetes

are larger and more variable, and are affected by factors missing here (medication, insulin resistance dynamics), so some gap is expected. Any real deployment would need subgroup-level accuracy targets.

Disagreement with Shen et al. They concluded that personalised training data was not required, whereas here it was the largest single gain. Three explanations fit:

1. A 45-person cohort learns population patterns less well, so personal data has more to compensate for.
2. CGMacros spans healthy to type 2 diabetes, so between-person differences are larger than within one diagnostic group.
3. Personal data here comes from the same 10-day window as the test meals, which may favour it.

This question is worth testing on a larger cohort with longer follow-up.

Practical meaning. With AUROC around 0.87–0.89 and about two-thirds of spikes detected, the model could flag meals likely to cause a large spike for an educational tool. It is not reliable enough for clinical decisions such as medication dosing.

7. Limitations and Ethics

Limitations

- **Small cohort.** 45 participants (14 with type 2 diabetes) supports a pilot, not firm conclusions; sex and age results are exploratory.
- **Noisy inputs.** Meals are self-logged estimates in free-living conditions, which caps achievable accuracy.
- **Short follow-up.** About 10 days per person, so personalisation adapts on roughly 5 days; menstrual, seasonal, medication and stress effects are not captured.
- **Missing factors.** No sleep, medication, food-order or meal-duration data.
- **Population.** US adults, mostly Hispanic/Latino; no external validation cohort.
- **Sensor dependence.** Libre and Dexcom disagree in absolute values, though relative findings held for both.
- **Selection.** Excluding overlapping meals under-represents back-to-back eating.
- **Ensemble choice.** The best AUROC was picked from three combinations.

Ethics

- De-identified public data used under its licence (attribution, non-commercial, share-alike); raw data is not redistributed.

- Not a medical device and not for clinical use; predictions should not guide diet or medication decisions.
- The remaining accuracy gap for type 2 diabetes is a fairness concern: a tool that works worst for higher-risk users could widen health inequalities. Deployment should require acceptable accuracy in every group served.
- Glucose and diet data are sensitive. Any real use needs consent, data minimisation and secure storage.

8. Conclusion and Future Work

Personal and meal-level data predicted individual glucose responses much better than carbohydrate content. Learning from a person's own earlier meals was the most effective step, cutting 60-minute error by 38% and reaching r = 0.78. Gradient-boosted trees and a temporal CNN were complementary, and their average gave the best spike detection (AUROC about 0.89). Personalisation helped type 2 diabetes participants most, but did not fully close their accuracy gap.

Answers to the research questions:

1. **RQ1:** Yes. LightGBM beat carbs only by 14% without history and 24% with it.
2. **RQ2:** Yes. Personalisation gave the largest single gain.
3. **RQ3:** Partly. The CNN was better at 30 minutes and for full curves; LightGBM was better at 60 and 120 minutes.
4. **RQ4:** Partly. All groups improved, but type 2 diabetes error stayed higher.

Future work:

- External validation on a larger CGM cohort with meal logs.
- Medication and sleep data, to target the type 2 diabetes gap.
- Nested cross-validation with hyperparameter tuning.
- Per-prediction uncertainty (e.g. conformal intervals) for safer use.
- Testing how many days of personal data are needed for the gain.

Code Availability and AI Use

Code, results and project documents: github.com/chrysl7076/glycemic-response-predictor. Every result can be regenerated from the public dataset with four commands listed in the README.

An AI assistant (Claude) helped write code and draft text. I defined the research questions, made the scoping and analysis decisions, reviewed and understood all code, and interpreted the results. My course instructor is aware of this use.

References

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